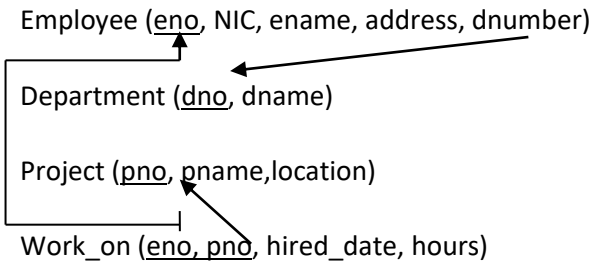
BSc (Hons) in Information Technology – Year 1

Lab Sheet 13 - Revision

IT1090 – Information Systems and Data Modeling

Consider the following relational schema.



1. Considering the following data types for the above schema, create the following relational database using CREATE TABLE SQL statement. Ensure that appropriate referential integrity constraints (foreign keys) are met. Save the script as "**CreateDB.sql**"

2. Employee(eno:INTEGER,NIC:CHAR(10),ename:VARCHAR(50),address:VARCHAR(50), dnumber:INTEGER)

   Department (dno:INTEGER,dname:varchar(50))

   Project (pno:INTEGER,pname:VARCHAR(20))

   Work_on(eno:INTEGER,pno:INTEGER,hired_date:DATATIME, hours:INTEGER)

3. Delete all the tables you have created using DROP TABLE SQL statement.

4. The following constraints exist for the schema. Use appropriate constraint mechanisms (such as CHECK constraint, etc.) to enforce the rules stated by modifying existing database schema and re-create the tables.

   - Ensure that the Employee's number is a three digit (0-9) number
   - Ensure that the Employee's NIC number contains 9 digits (0-9) and one character which is "V" or "v"
   - Ensure that all the Employees have a name.
   - Ensure that the number of working hours is a positive value. (> 0)
   - In the Department table, the value for dname attribute must be one of the following: "Headquarters", "Administration", or "Research".

5. Insert the following data using the INSERT SQL statement

Employee

Ename	NIC	Eno	address	dnumber
Smith,John B	901250087V	100	731 Fondren, Houston	5
Wrong, Franklin T.	895290452V	101	638 Voss, Houston	5
Zelaya, Alicia J.	923859070V	102	33321 Castle, Spring	4
Wallace, Jennifer S.	918257988V	103	291 Berry, Bellaria	4
Narayan, Remesh K.	904850045V	104	975 Fire Oak, Humble	5
English, Joyce A.	889950600V	105	5631 Rice, Houston	1
Jabbar, Ahamad V	915850123V	106	980 Dallas, Houston	1

Department

dno	dname
1	Headquarters
4	Administration
5	Research

Project

pno	pname	Location
1	Road Construction	Bellaire
2	Power plant	Sugarland
3	Reproduction	Houston

4	Recycling	Stafford

Work_on

eno	pno	hired_date	hours
100	1	12/04/2011	12
100	2	25/02/2011	1
100	3	15/05/2011	20
101	1	02/06/2011	50
102	4	08/06/2011	5
103	3	12/07/2011	3
104	2	20/07/2011	40
105	3	18/05/2011	1
105	1	01/07/2011	11
106	4	27/06/2011	1

6. Delete "***Wallace, Jennifer S.***" from the Employee table.
7. Change the name of the project **"Recycling**" to **"Water Recycling".**